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RESEARCH ARTICLE

Relative Comparison of the Static Magnetic Field and Oscillating Magnetic Field Effects on Gram-Negative and Gram-Positive Bacteria via Evaluating Oxidative Stress and DNA Fragmentation

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ABSTRACT

The effects of static magnetic field (SMF) and oscillating magnetic field (OMF) on oxidative stress responses and DNA fragmentation were investigated in *Staphylococcus aureus* and *Escherichia coli*. Bacterial cultures were exposed to magnetic field intensities ranging from 2 to 16 mT at 50 Hz for 8 h. Exposure to both SMF and OMF caused intensity-dependent decreases in optical density, indicating growth inhibition. In parallel, nitric oxide and superoxide anion radical levels, as well as the activities of catalase and superoxide dismutase, increased significantly in a field-strength–dependent manner. Glutathione disulfide levels were enhanced while reduced glutathione values decreased, reflecting a redox imbalance. Moreover, malondialdehyde and DNA fragmentation levels were markedly elevated, confirming oxidative and genotoxic stress. Comparative analysis revealed that *E. coli* exhibited more pronounced responses than *S. aureus*, with significant effects emerging above approximately 6–8 mT for OMF and 8–10 mT for SMF exposure. Overall, these findings demonstrate that both OMF and SMF can induce oxidative stress and DNA damage in bacterial cells, with OMF producing stronger effects. The results highlight the differential susceptibility of Gram-positive and Gram-negative bacteria to electromagnetic exposure and suggest potential implications for non-antibiotic antimicrobial strategies based on magnetic fields. Bioelectromagnetics. 00:00–00, 2026. © 2026 © 2026 Bioelectromagnetics Society.

1 | Introduction

Due to the increase in deaths by bacterial infections, new and effective methods need to be developed. Thanks to these methods, the increase in resistance to existing antibiotics can be prevented (von Bubnoff 2006). Antibiotics may be grouped, that is, as bactericidal and bacteriostatic according to their targets in the cell. Bacteriostatic antibiotics specifically inhibit ribosomal functions (Lambert 2012), while bactericidal ones act on translation process (Ajiboye et al. 2016), synthesis of cell wall, DNA repair, and replication (Kohanski et al. 2007). There is also evidence to suggest that oxidative stress also plays a role (Mutagenesis et al. 2010). It is also known that some antibacterial products cause redox imbalance by affecting antioxidant enzyme activities (Samoilova et al. 2014). For this reason, the effectiveness of the magnetic field effect on bacteria through these parameters has attracted our attention.

Living organisms, especially microorganisms, are highly sensitive to environmental factors, such as magnetic fields, radiation, frequency, electric field, heat, and so forth. Magnetic fields are

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Summary

- Low-intensity OMF and SMF induced oxidative stress in *S. aureus* and *E. coli* cultures.
- DNA fragmentation and glutathione depletion were observed in treated bacterial cells.
- OMF were found to be more effective than SMF in inducing bacterial damage.

among these factors and can affect the behavior of bacteria as a biological systems. The simplicity of structure of bacteria are affected by the magnetic field due to electron transfer, ion transporting across the membrane, and metabolic activities. Recent studies have shown that magnetic fields can have varying degrees of biological effects on both Gram-positive and Gram-negative bacteria. In this work, it is shown that intensity of the magnetic field (direct or alternating) and duration exposure play critical roles in bacterial growth rate, metabolic activity, cell membrane permeability, and DNA integrity. Biostimulation effects can be observed in low static-intensity magnetic fields, while oscillating magnetic field (OMF) and high intensity magnetic field cause to oxidative stress and cellular damage (Fojt et al. 2004; Segatore et al. 2012; Gaafar et al. 2008; Pang and Jalbout 2004).

Pang et al. concluded that closed hydrogen-bonded molecular chains still exist in water after the first-order phase transition from ice crystals to water (Pang and Shen 2013). It has been shown that the effect of the magnetic fields on the growth and bacterial viability depend on many factors such as magnetic field intensity, time exposure, and the type of organism. Many researchers have shown that the exposure of bacteria to magnetic fields, can significantly affect the viability (Galvanovskis and Sandblom 1998). In the literature, it has been revealed that the reason why the different groups of bacteria (Gram negative and Gram positive) exposed to a magnetic field for a long time are different in the case of magnetic field excitation or exposure to magnetic field is due to the effect of the cell wall transport mechanism and the cell membrane structures of these groups are different (Konopacki and Rakoczy 2019). The effect of magnetic field on bacterial growth was also studied (Fijałkowski et al. 2013; Nawrotek et al. 2014).

The aim of this study is to comparatively investigate the effects of oscillating and static magnetic fields (SMFs) on oxidative stress and DNA damage in gram-positive and Gram-negative bacteria, in response to the growing need for alternative antimicrobial strategies due to rising antibiotic resistance. There are few studies on bacteria and magnetic fields, but the mechanisms that cause bacterial death are not mentioned in them. So, in this study, we analyzed the influence of SMF and OMF on the fundamental survival parameters of bacterias. It was specified to analyse the oscillating and SMFs influence on the Gram-negative and Gram-positive bacteria, which have different cell wall structure. The effects of magnetic fields on the cells was analysed via the measuring of the metabolic activity and growth ratio of bacteria.

2 | Materials and Methods

In this study, we obtain the oscillating and SMFs from power supply (PHYWE Constant/Netzgerät Universal Power Supply) (Figure 1). To generate the required magnetic fields, the solenoid was energized with either direct current (DC) to produce SMFs and alternating current (AC) to produce OMFs. The electrical current (I) was precisely controlled using a series-connected sliding rheostat with a maximum resistance of 10 Ω and a maximum current capacity of 5 A. The solenoid, which serves as the core of the exposure system, was wound using enameled copper wire (approximate diameter of 1.0 mm) to form a total of 1000 turns (N). The current passing through the solenoid was monitored using a series-connected digital Multimeter (TT T-ECHNI-C VC9804A multimeter), which has a maximum measurable current range of 20 A.

The SMF intensity is calculated by the formula $B = \mu_0 nI$, where B is the magnetic field intensity (Tesla, T), μ_0 is the magnetic permeability of vacuum ($4\pi \times 10^{-7}$ H/m), n is the number of turns per unit length ($n = N/L$) and I is the direct current (A) passing through the coil. In this setup, the solenoid length (L) was 0.20 m with $N = 1000$ turns. We controlled B by adjusting the amplitude of the current I. In this study, the OMF intensity is expressed as $B(t) = \mu_0 n I_0 \sin(\omega t)$, where $B(t)$ is the time-varying magnetic field (T), I_0 is the maximum current (A), $\omega = 2\pi f$ is the angular frequency (s^{-1}) with a frequency $f = 50$ Hz, and t represents time (s).

Experiments were carried out at 2–16 mT static and OMF values. All experiments were repeated independently 3 times. In addition within each experiment, three experimental replicates were either exposed to magnetic field or unexposed control groups. Therefore, altogether 9 experimental samples were monitored in every experimental setup.

2.1 | Chemicals, Bacterial Strains, and Cell Growth

All chemicals used in this study were obtained from Sigma-Aldrich. *S. aureus* (ATCC 29213) and *E. coli* (ATCC 25922) were propagated on a lysogeny broth at 37°C. The cell growth values were analysed at 600 nm wavelength measuring the optical density (OD) by spectrophotometric reader after 8 h.

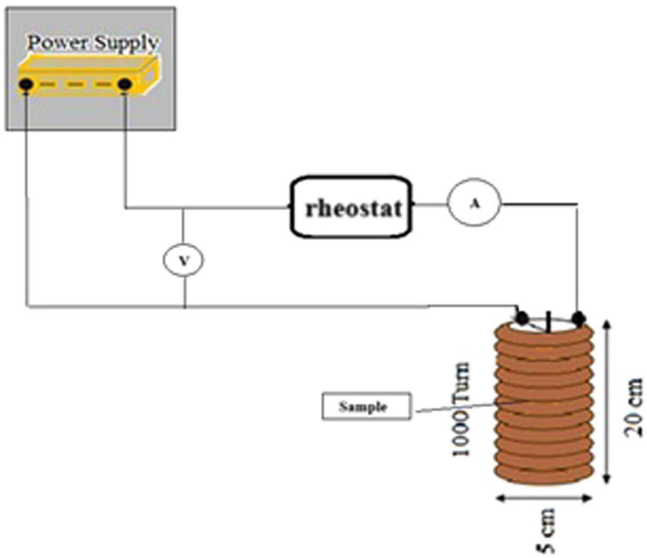

FIGURE 1 | Schematic diagram of static magnetic field and oscillating magnetic field apparatus.

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TABLE 1 | Results showing the effects of magnetic field type and intensity on all measured parameters.

Parameter	Species	Field type (OMF vs. SMF)	Field intensity	Field × Intensity
OD600	*E. coli*	$p = 0.004$	$p < 0.001$	$p = 0.017$
OD600	*S. aureus*	$p = 0.032$	$p = 0.002$	$p = 0.094$
Superoxide anion	*E. coli*	$p < 0.001$	$p < 0.001$	$p = 0.022$
Superoxide anion	*S. aureus*	$p = 0.018$	$p = 0.004$	$p = 0.112$
Nitric oxide	*E. coli*	$p = 0.002$	$p < 0.001$	$p = 0.028$
Nitric oxide	*S. aureus*	$p = 0.068$	$p = 0.006$	$p = 0.138$
SOD activity	*E. coli*	$p = 0.009$	$p < 0.001$	$p = 0.034$
SOD activity	*S. aureus*	$p = 0.081$	$p = 0.012$	$p = 0.156$
Catalase activity	*E. coli*	$p = 0.012$	$p < 0.001$	$p = 0.042$
Catalase activity	*S. aureus*	$p = 0.094$	$p = 0.018$	$p = 0.173$
GSH level	*E. coli*	$p = 0.006$	$p < 0.001$	$p = 0.028$
GSH level	*S. aureus*	$p = 0.087$	$p = 0.021$	$p = 0.149$
GSSG level	*E. coli*	$p = 0.005$	$p < 0.001$	$p = 0.024$
GSSG level	*S. aureus*	$p = 0.074$	$p = 0.019$	$p = 0.132$
MDA	*E. coli*	$p < 0.001$	$p < 0.001$	$p = 0.018$
MDA	*S. aureus*	$p = 0.023$	$p = 0.003$	$p = 0.091$
DNA fragmentation	*E. coli*	$p = 0.003$	$p < 0.001$	$p = 0.031$
DNA fragmentation	*S. aureus*	$p = 0.082$	$p = 0.014$	$p = 0.158$

Note: Statistical significance was accepted at $p < 0.05$. Two-way ANOVA was used to evaluate the main effects of magnetic field type (OMF vs. SMF), magnetic field intensity, and their interaction.
Abbreviations: ANOVA, analysis of variance; OMF, oscillating magnetic field; SMF, static magnetic fields.

2.2 | Measurement of Oxidative Stress Values

Superoxide anion values of bacteria-free extract were measured by the method of Becerra and Albesa (Becerra and Albesa 2002). 0.5 mL bacteria-free extract and 2.5 mL nitroblue tetrazolium were incubated at 37°C for 30 min. Then 0.1 M hydrochloric acid (HCl) was added and read at 575 nm. The nitric oxide value was analysed by the colorimetric method based on the Griess reaction (Green et al. 1982).

The activity of total superoxide dismutase (SOD) enzyme was measured by analyse of pyrogallol autoxidation and illumination at 440 nm wavelength by Marklund and Marklund's procedure (Marklund and Marklund 1974). For measurement of catalase (CAT) activity homogenate (50 mL), phosphate buffer (pH 7.0; 2 mL), and hydrogen peroxide (H_2O_2) (1 mL) were added in a cuvette and read at 240 nm using a spectrophotometer (Aebi 1984). The level of malondialdehyde (MDA) was measured by reacting with thiobarbituric acid (TBA) at the wavelength of 532 nm because it is a marker for lipid peroxidation (Ohkawa et al. 1979). The glutathione (GSH) levels were determined by Ellman's procedure (Ellman 1959). Cell-free extract (1.0 mL) and 25% trichloroacetic acid (TCA) (0.1 mL) were centrifuged for 10 min and 0.1 mL supernatant was added to 2 mL 5,5′-Dithiobis (2-nitrobenzoic acid) (DTNB), which was prepared in sodium phosphate buffer (0.2 M; pH 8.0) and read at 412 nm. The level of glutathione disulfide (GSSG) was measured by the method of Hissin and Hilf (Hissin and Hilf 1976). 50 mL of bacteria-free extract and 20 mL of N-ethylmaleimide (0.04 M) were mixed for preventing oxidation of GSH to GSSG, then incubated at 25°C for 30 min and 1.68 mL of disodium hydrogen phosphate (Na_2HPO_4) (0.3 M) was added to it followed by DTNB reagent (250 mL) and read at 412 nm. Each wavelength was selected based on the specific absorbance maximum of the reaction product generated in each biochemical assay, in accordance with well-established and widely accepted analytical protocols. These wavelengths correspond to the highest sensitivity and specificity for detecting the target molecules.

2.3 | Measurement of Fragmented DNA Values

The fragmented DNA was measured by Burton's procedure (Burton 1956). Bacteria free extract was centrifuged at 4°C for 15 min. Then supernatant was separated and treated with 10% 1.50 mL trichloroacetic acid. Also, the pellet was treated with 5% 0.65 mL trichloroacetic acid too. The mixtures were allowed to precipitate overnight at 4°C, then centrifuged for 10 min. The mixtures were boiled (100°C) and cooled to 25°C and centrifuged again for 5 min. 0.5 mL of supernatant was treated with 1 mL diphenylamine reagent, incubated at 37°C for 4 h and read at 600 nm.

2.4 | Statistics

All statistical calculations were conducted using the IBM SPSS Statistics Program, version 26.0 for Windows (SPSS Inc., USA). Statistical differences between groups (control group without magnetic field and groups exposed to different magnetic field values) for measured parameters were analyzed using one-way ANOVA followed by Tukey's post hoc test for multiple comparisons. All findings were represented as the means ± standard deviation (SD) and statistically significant was accepted at $p < 0.05$. For parameter comparisons between *Escherichia coli* and *Staphylococcus aureus* were conducted at each magnetic

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field intensity using independent samples *t*-tests. The normality of data distribution was confirmed using the Shapiro–Wilk test. In cases where the assumption of normality was violated, nonparametric Mann–Whitney *U* tests were applied. The effects of magnetic field type and intensity were analyzed by two-way ANOVA, followed by Tukey's post hoc test for multiple comparisons among magnetic field intensities within each field type and bacterial species. A $p < 0.05$ was considered statistically significant.

3 | Results

The results were evaluated across eight different magnetic field intensities (2, 4, 6, 8, 10, 12, 14, and 16 mT) in addition to the control group for both OMF and SMF exposures. A gradual, intensity-dependent change was observed in all measured parameters. For *E. coli*, significant changes in parameters begin around 6–8 mT under OMF exposure and 8–10 mT under SMF exposure. For *S. aureus*, comparable effects appear at slightly higher intensities (8–10 mT for OMF and 10–12 mT for SMF).

Analyses revealed significant main effects of magnetic field type (OMF vs. SMF) and magnetic field intensity on bacterial growth and oxidative stress–related parameters (Table 1). Overall, OMF exposure induced stronger effects than SMF, particularly in *E. coli*, while increasing the magnetic field intensity produced a dose-dependent response in both bacterial species. A significant field type × intensity interaction was observed mainly in *E. coli*, indicating that the difference between OMF and SMF increased with higher field intensities. In contrast, although *S. aureus* was affected by magnetic field exposure, the relative difference between OMF and SMF remained largely constant across all intensities. Post hoc analysis demonstrated significant intensity-dependent effects of magnetic field exposure on multiple bacterial parameters (Tables 2 and 3). These effects were more pronounced in *E. coli*, particularly under OMF exposure, whereas *S. aureus* exhibited comparatively modest and more uniform responses across increasing field intensities.

3.1 | Cell Growth

The obtained results about the bacteria growth are presented in Figure 2. There was a decrease in cell growth with increasing magnetic field intensity in both oscillating and SMFs. But OMF was more effective than SMF. Also, *E. coli* was more affected than *S. aureus*. Under both OMF and SMF, the OD of *E. coli* decreased more sharply than that of *S. aureus* with increasing field intensity. Statistical comparison between the two species showed no significant differences at lower intensities (≤ 6 mT), whereas at 8 mT and above, *E. coli* exhibited significantly lower OD values than *S. aureus* ($p < 0.05$). This trend was more pronounced under OMF exposure, suggesting that *E. coli* is more sensitive to alternating magnetic fields in terms of growth inhibition. The numerical decrease in OD reflects a biologically relevant inhibition of bacterial proliferation. A reduction in OD after 8 h corresponds to a substantial decrease in viable bacterial count due to impaired metabolic activity. Moreover, these changes were consistent with increased oxidative stress parameters (Figures 3–9), indicating that the observed OD reduction represents a meaningful physiological response rather than random variation.

3.2 | Superoxide Anion Radical Level

For *E. coli* there was not differences between control, 2 and 4 mT groups; for *S. aureus* there was not differences between control, 2, 4, and 6 mT groups. The level of superoxide anion radical of *E. coli* increased significantly at 6 mT (OMF) and 8 mT (SMF) ($p < 0.05$) and *S. aureus* increased at 8 mT (OMF) and 10 mT (SMF). OMF was more effective than SMF. Also, *E. coli* was more affected than *S. aureus* (Figure 3).

3.3 | Nitric Oxide Level

The level of nitric oxide (NO) significantly increased ($p < 0.05$) by increasing magnetic field intensity (starting from 6 mT [OMF], 8 mT [SMF] for *E. coli* and 10 mT [OMF], 12 mT [SMF] for *S. aureus*) when compared to the control (Figure 4).

3.4 | Superoxide Dismutase and Catalase Activities

The activities of superoxide dismutase (Figure 5) and catalase (Figure 6) in the bacterial cells treated with both oscillating and SMFs increased significantly ($p < 0.05$). But OMF was more effective than SMF. Also, *E. coli* was more affected than *S. aureus*.

3.5 | GSH and GSSG Levels

GSH level (Figure 7) in cells of *S. aureus* and *E. coli* decrease significantly ($p < 0.05$) following OMF and SMF treatment in intensity dependent manner. Conversely, OMF and SMF significantly ($p < 0.05$) increased the level of GSSG (Figure 8) in *S. aureus* and *E. coli*.

3.6 | Malondialdehyde Level

For *E. coli* there was not differences between control, 2 and 4 mT groups; for *S. aureus* there was not differences between control, 2, 4, and 6 mT groups. MDA, a biomarker of lipid peroxidation, increased significantly in *E. coli* and *S. aureus* cells treated with oscillating and SMFs (Figure 9).

3.7 | Fragmented DNA Concentration

For *E. coli* there was not differences between control, 2, 4, 6, and 8 mT treated groups; for *S. aureus* there was not differences between control, 2, 4, 6, 8, and 10 mT treated groups. There was intensity-dependent increase in fragmented DNA (%) following treatment of *E. coli* and *S. aureus* with OMF and SMF (Figure 10).

4 | Discussion

In this study, the ability of Gram-negative *E. coli* and Gram-positive *S. aureus* bacteria species, both pathogenic, to be affected by different levels of magnetic field are compared.

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TABLE 2 | Effects of different magnetic field intensities on bacterial growth (*E. coli*), membrane damage, oxidative stress, and genotoxicity parameters under OMF and SMF exposure.

Intensity (mT)	OD600 (OMF)	OD600 (SMF)	GSH (OMF)	GSH (SMF)	MDA (OMF)	MDA (SMF)	SOD (OMF)	SOD (SMF)	CAT (OMF)
Control	0.4 ± 0.02^{a}	0.4 ± 0.016^{a}	0.9 ± 0.06^{a}	0.85 ± 0.1^{a}	0.3 ± 0.05^{a}	0.4 ± 0.06^{a}	4.7 ± 0.51^{a}	4.2 ± 1.1^{a}	2.68 ± 1.06^{a}
2	0.39 ± 0.014^{a}	0.395 ± 0.017^{a}	0.82 ± 0.04^{a}	0.78 ± 0.06^{a}	0.35 ± 0.12^{a}	0.42 ± 0.09^{a}	5.15 ± 1.0^{a}	$5.04 \pm 0{,}81^{b}$	3.15 ± 078^{b}
4	0.375 ± 0.016^{a}	0.389 ± 0.014^{a}	0.78 ± 0.08^{a}	0.74 ± 0.03^{a}	0.44 ± 0.11^{b}	0.47 ± 0.05^{a}	5.34 ± 1.31^{a}	5.23 ± 0.8^{b}	3.35 ± 0.76^{b}
6	0.345 ± 0.013^{b}	0.374 ± 0.018^{a}	0.75 ± 0.06^{b}	0.7 ± 0.04^{b}	0.82 ± 0.13^{c}	0.5 ± 0.08^{a}	8.1 ± 0.52^{b}	5.9 ± 1.43^{c}	7.1 ± 1.25^{c}
8	0.31 ± 0.012^{c}	0.355 ± 0.014^{b}	0.54 ± 0.03^{c}	0.66 ± 0.06^{b}	1.34 ± 0.06^{d}	0.83 ± 0.07^{b}	11.7 ± 0.91^{c}	9.2 ± 0.99^{d}	9.7 ± 0.81^{d}
10	0.26 ± 0.014^{d}	0.31 ± 0.013^{c}	0.35 ± 0.06^{d}	0.51 ± 0.04^{c}	1.74 ± 0.16^{e}	1.14 ± 0.13^{c}	15.6 ± 1.3^{d}	13.06 ± 1.01^{e}	13.6 ± 0.85^{e}
12	0.195 ± 0.012^{e}	0.25 ± 0.001^{d}	0.22 ± 0.03^{e}	0.27 ± 0.02^{d}	1.97 ± 0.018^{f}	1.49 ± 0.06^{d}	18.11 ± 0.72^{e}	15.54 ± 1.92^{e}	16.12 ± 1.53^{f}
14	0.115 ± 0.01^{f}	0.184 ± 0.012^{e}	0.16 ± 0.02^{f}	0.21 ± 0.03^{e}	2.08 ± 0.13^{g}	1.73 ± 0.15^{e}	21.1 ± 2.02^{f}	18.22 ± 1.04^{f}	19.71 ± 0.97^{g}
16	0.06 ± 0.01^{g}	0.13 ± 0.013^{f}	0.06 ± 0.02^{g}	0.13 ± 0.02^{f}	2.3 ± 0.14^{g}	1.89 ± 0.11^{f}	25.5 ± 1.01^{g}	21.55 ± 1.5^{g}	22.5 ± 1.61^{h}

	CAT (SMF)	DNA frag. (OMF)	DNA frag. (SMF)	GSSG (OMF)	GSSG (SMF)	Superoxide anion radical (OMF)	Superoxide anion radical (SMF)	Nitric oxide (OMF)	Nitric oxide (SMF)
Control	2.2 ± 0.62^{a}	2.7 ± 1.71^{a}	2.21 ± 0.68^{a}	0.3 ± 0.06^{a}	0.39 ± 0.08^{a}	0.7 ± 0.3^{a}	0.8 ± 0.17^{a}	1.71 ± 0.31^{a}	1.8 ± 0.29^{a}
2	3.05 ± 1.07^{a}	3.15 ± 0.78^{a}	3.07 ± 1.07^{a}	0.35 ± 0.13^{a}	0.41 ± 0.1^{a}	1.15 ± 0.38^{b}	1.05 ± 0.3^{a}	2.14 ± 0.42^{b}	2.04 ± 0.65^{a}
4	3.22 ± 1.28^{a}	3.34 ± 0.8^{a}	3.23 ± 1.24^{a}	0.44 ± 0.12^{a}	0.46 ± 0.07^{a}	1.34 ± 0.32^{b}	1.22 ± 0.78^{a}	2.33 ± 0.09^{b}	2.19 ± 0.5^{a}
6	5.2 ± 0.82^{b}	5 ± 1.69^{b}	4.21 ± 1.38^{a}	0.48 ± 0.14^{a}	0.51 ± 0.13^{a}	2.1 ± 0.24^{c}	1.44 ± 0.14^{a}	3.12 ± 0.08^{c}	2.43 ± 0.6^{b}
8	6 ± 095^{c}	5.7 ± 1.19^{b}	4.6 ± 1.56^{a}	0.9 ± 0.08^{b}	0.53 ± 0.09^{a}	27 ± 0.29^{d}	2.2 ± 0.4^{b}	3.71 ± 0.48^{c}	3.21 ± 0.27^{c}
10	11.05 ± 1.24^{d}	12.6 ± 0.91^{c}	5.9 ± 2.04^{b}	1.34 ± 0.17^{c}	0.9 ± 0.14^{b}	3.6 ± 0.43^{e}	3.05 ± 0.23^{c}	4.59 ± 0.28^{d}	4.06 ± 0.38^{d}
12	13.55 ± 0.78^{e}	17.11 ± 1.84^{d}	11.54 ± 0.8^{c}	1.57 ± 0.13^{d}	1.15 ± 0.08^{c}	4.11 ± 0.29^{f}	3.55 ± 0.52^{d}	5.12 ± 0.12^{e}	4.54 ± 0.56^{d}
14	16.22 ± 1.54^{f}	20.7 ± 1.01^{e}	15.21 ± 1.58^{d}	1.7 ± 0.14^{e}	1.38 ± 0.16^{d}	4.8 ± 0.31^{g}	4.22 ± 0.25^{e}	$57 \pm 0{,}45^{f}$	5.25 ± 0.4^{e}
16	20.54 ± 1.04^{g}	24.51 ± 1.67^{f}	19.56 ± 1.08^{e}	$1.9 \pm 0{,}12^{f}$	1.74 ± 0.13^{e}	5.5 ± 0.29^{h}	4.64 ± 0.32^{e}	6.52 ± 0.19^{g}	5.55 ± 0.61^{e}

Note: Values are mean ± SD. Significance at $p < 0.05$. Differences among groups were analyzed by one-way ANOVA followed by Tukey's post hoc test. Within each column, means superscript with different letters (a–h) are significantly different.

Abbreviations: CAT, catalase; GSH, glutathione; MDA, malondialdehyde; OMF, oscillating magnetic field; SMF, static magnetic fields; SOD, superoxide dismutase.

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TABLE 3 | Effects of different magnetic field intensities on bacterial growth (*S. aureus*), membrane damage, oxidative stress, and genotoxicity parameters under OMF and SMF exposure.

Intensity (mT)	OD600 (OMF)	OD600 (SMF)	GSH (OMF)	GSH (SMF)	MDA (OMF)	MDA (SMF)	SOD (OMF)	SOD (SMF)	CAT (OMF)
Control	0.4 ± 0.017^{a}	0.4 ± 0.016^{a}	$0.81 \pm 0{,}05^{a}$	0.75 ± 0.08^{a}	0.41 ± 0.13^{a}	0.44 ± 0.09^{a}	5.33 ± 1.11^{a}	5.55 ± 0.58^{a}	4.33 ± 0.6^{a}
2	0.394 ± 0.015^{a}	0.397 ± 0.013^{a}	0.75 ± 0.08^{a}	0.7 ± 0.04^{a}	0.47 ± 0.07^{a}	0.46 ± 0.13^{a}	6.77 ± 1.81^{a}	6.1 ± 0.91^{a}	5.77 ± 0.89^{a}
4	0.38 ± 0.016^{a}	0.39 ± 0.01^{a}	0.71 ± 0.02^{a}	0.67 ± 0.04^{a}	0.51 ± 0.15^{a}	0.49 ± 0.08^{a}	7.5 ± 0.59^{a}	$6.56 \pm 1{,}63^{a}$	6.2 ± 1.59^{a}
6	0.365 ± 0.016^{b}	0.382 ± 0.015^{a}	0.69 ± 0.03^{a}	0.64 ± 0.06^{a}	0.59 ± 0.11^{a}	0.51 ± 0.17^{a}	8.16 ± 1.0^{a}	7.2 ± 1.21^{a}	7.16 ± 1.15^{a}
8	0.33 ± 0.015^{c}	0.366 ± 0.013^{b}	0.65 ± 0.06^{b}	0.62 ± 0.04^{a}	0.98 ± 0.17^{b}	0.55 ± 0.07^{a}	13.34 ± 1.12^{b}	8.65 ± 1.53^{a}	11.34 ± 1.41^{b}
10	0.285 ± 0.012^{d}	0.328 ± 0.012^{c}	0.44 ± 0.03^{c}	0.56 ± 0.05^{b}	1.32 ± 0.14^{c}	0.88 ± 0.13^{b}	18.2 ± 1.41^{c}	14.45 ± 0.93^{b}	15.2 ± 0.86^{c}
12	0.224 ± 0.013^{e}	0.275 ± 0.01^{d}	0.34 ± 0.02^{d}	0.44 ± 0.03^{c}	1.55 ± 0.21^{d}	1.23 ± 0.16^{c}	22.8 ± 1.07^{d}	18.43 ± 1.61^{c}	19.8 ± 1.54^{d}
14	0.16 ± 0.011^{f}	0.204 ± 0.013^{e}	0.17 ± 0.02^{e}	0.31 ± 0.04^{d}	1.67 ± 0.13^{d}	1.4 ± 0.09^{d}	25.42 ± 2.03^{d}	22.31 ± 1.42^{d}	22.42 ± 1.35^{e}
16	0.095 ± 0.011^{g}	0.155 ± 0.011^{f}	0.11 ± 0.03^{f}	0.18 ± 0.02^{e}	1.8 ± 0.18^{e}	1.52 ± 0.19^{e}	31.94 ± 1.6^{e}	27 ± 1.82^{e}	27.94 ± 2.13^{f}

	CAT (SMF)	DNA frag. (OMF)	DNA frag. (SMF)	GSSG (OMF)	GSSG (SMF)	Superoxide anion radical (OMF)	Superoxide anion radical (SMF)	Nitric oxide (OMF)	Nitric oxide (SMF)
Control	4.55 ± 0.98^{a}	3.37 ± 1.49^{a}	3.67 ± 1.21^{a}	0.4 ± 0.1^{a}	0.42 ± 0.07^{a}	$1.13 \pm 0{,}2^{a}$	0.95 ± 0.33^{a}	1.34 ± 0.47^{a}	1.54 ± 0.29^{a}
2	5.1 ± 1.43^{a}	3.8 ± 0.78^{a}	4.11 ± 1.34^{a}	0.45 ± 0.04^{a}	0.47 ± 0.11^{a}	1.26 ± 0.28^{a}	1.2 ± 0.24^{a}	1.38 ± 0.65^{a}	1.5 ± 0.27^{a}
4	5.8 ± 0.54^{a}	4.21 ± 1.49^{a}	4.4 ± 0.45^{a}	0.52 ± 0.13^{a}	0.49 ± 0.06^{a}	1.4 ± 0.35^{a}	1.37 ± 0.19^{a}	1.49 ± 0.36^{a}	1.56 ± 0.48^{a}
6	6.6 ± 0.9^{a}	5 ± 1.08^{a}	4.51 ± 0.94^{a}	0.55 ± 0.09^{a}	0.52 ± 0.15^{a}	1.64 ± 0.28^{a}	1.51 ± 0.43^{a}	1.78 ± 0.34^{a}	1.78 ± 0.29^{a}
8	7.55 ± 1^{a}	6.35 ± 1.81^{a}	5.1 ± 0.99^{a}	0.71 ± 0.15^{a}	0.56 ± 0.05^{a}	2.24 ± 0.36^{b}	1.69 ± 0.36^{a}	2.35 ± 0.38^{b}	2.1 ± 0.34^{b}
10	11.45 ± 1.35^{b}	6.69 ± 1.93^{a}	5.78 ± 1.1^{a}	1.02 ± 0.12^{b}	0.69 ± 0.11^{a}	3.1 ± 0.55^{c}	2.64 ± 0.37^{b}	3.21 ± 0.36^{c}	2.46 ± 0.66^{b}
12	14.43 ± 0.92^{c}	10.78 ± 1.48^{b}	7.07 ± 1.56^{b}	1.26 ± 0.1^{c}	1.06 ± 0.12^{b}	3.7 ± 0.21^{c}	3.32 ± 0.45^{b}	3.82 ± 0.41^{c}	3.44 ± 0.37^{c}
14	19.01 ± 1.81^{d}	16.44 ± 1.32^{c}	12.04 ± 1.71^{b}	1.42 ± 0.07^{d}	1.2 ± 0.12^{c}	4.33 ± 0.28^{d}	3.93 ± 0.26^{c}	4.44 ± 0.51^{d}	4.02 ± 0.33^{c}
16	24.9 ± 1.7^{e}	19.96 ± 2.07^{d}	15.34 ± 1.38^{c}	1.58 ± 0.13^{d}	1.42 ± 0.11^{c}	4.86 ± 0.44^{d}	4.2 ± 0.36^{c}	4.95 ± 0.37^{d}	4.33 ± 0.64^{c}

Note: Values are mean ± SD. Significance at $p < 0.05$. Differences among groups were analyzed by one-way ANOVA followed by Tukey's post hoc test. Within each column, means superscript with different letters (a–g) are significantly different.

Abbreviations: CAT, catalase; GSH, glutathione; MDA, malondialdehyde; OMF, oscillating magnetic field; SMF, static magnetic fields; SOD, superoxide dismutase.

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When we look at the effects of applied magnetic fields on cell proliferation, we see that the effect of OMF on bacteria is greater. In addition, *E. coli* was more affected by the magnetic field at the same level than *S. aureus*. The changes in OD represent biologically meaningful growth inhibition when interpreted within the context of bacterial proliferation kinetics. In logarithmic growth phase, decrease in OD corresponds to a considerable reduction in viable cell count and metabolic activity (Stevenson et al. 2016). The observed decreases in OD were also consistent with increased oxidative stress parameters and DNA fragmentation, confirming that the growth suppression is accompanied by significant physiological stress rather than random fluctuation. These statistically significant reductions in OD suggest impaired bacterial proliferation, likely involving disruption of cell division and essential metabolic processes. Changes in growth rate inferred from OD measurements provide important information about cellular integrity and metabolic activity. Although bacterial morphology was not directly examined in the present study, growth inhibition is commonly associated with underlying morphological and structural alterations, including membrane damage, increased permeability, and oxidative injury to cellular components. The combined evidence from growth suppression, oxidative stress induction, and membrane lipid peroxidation strongly implies magnetic field–induced stress at the cellular level. In previous studies, the effect of magnetic field on gram-positive bacteria was compared with control groups by applying it for a longer time, and it was stated that bacterial growth and metabolic activity could be explained by micro-mixing events. Here, they tried to explain that ions such as Ca^{2+} and K^{+} in the bacterial environment cause changes in cell proliferation by acting with the magnetic field (Hristov 2010). RMF interacts with various charges particles, for example, ions produce eddy currents in the culture medium. Eddy currents can produce local MFs around the ions, in combination with externally applied MF, causing their rotation to be induced and thus the fluid to move relative to the MF vector. The rotating ions as a result of this process form systems that cause the micro mixing effect. Therefore, the RMF generator can be considered a micro mixer. Micro stirrers using magnetic forces represent an important class of mixing. This type of mixer uses the Lorentz force to agitate liquids and induce secondary complex chaotic flows (Gaafar et al. 2008; Anton-Leberre et al. 2010; Ryu et al. 2004). Due to the magnetic field, change permeability of the cell membrane, cause biological effects in the organism and free radicals that are released (Galvanovskis and Sandblom 1998). This is thought to have effects on cell proliferation. The biological effects of magnetic field exposure observed in this study were primarily reflected through measurable changes in bacterial growth, oxidative stress markers, antioxidant defense responses, and DNA integrity. Specifically, growth inhibition assessed by OD measurements, increased levels of superoxide anion and nitric oxide, altered activities of antioxidant enzymes (superoxide dismutase and catalase), disruption of the glutathione redox balance (GSH and GSSG), elevated lipid peroxidation (MDA), and increased DNA fragmentation collectively constituted the key parameters. The intensity-dependent alteration of these parameters indicates that magnetic field exposure induces a oxidative and genotoxic stress response.

The applied external magnetic field affected the Lande g factor of the bacterial samples placed in the middle of the solonoid. When the bacteria is exposed to a magnetic field, it can cause the total angular moments of the electrons in the cells to change. It is inevitable that the total angular moments depending on the L orbital angular momentum and S spin angular momentum will change. This leads to shifts and splits in the Landé g factor (Binhi and Rubin 2007). These splits and shifts occur in bacteria; it creates intracellular induced currents, which affect the cell membrane, proteins and DNA by increasing the production of reactive oxygen species (ROS) (Sadri et al. 2017). It affects the ion channels and membrane potential in the bacterial cell membrane, and can affect magnetic field-sensitive molecules such as Fe-S proteins and cytochrome complexes on the electron transport chain (Kiley and Beinert 2003). This may damage metabolic activities and DNA structure. In this study, these changes are presented in detail in figures.

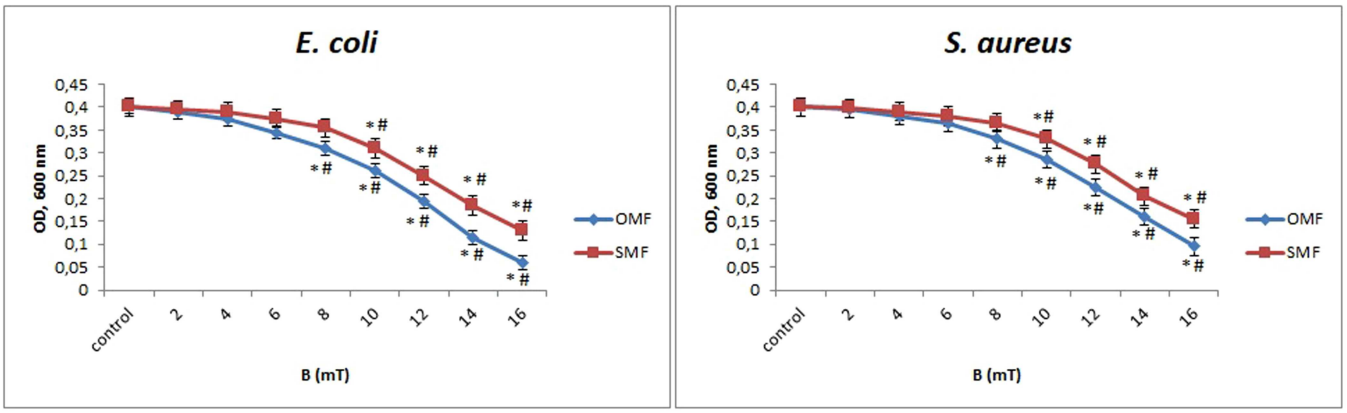

FIGURE 2 | OMF and SMF effects on cell growth of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Error bars represent standard deviation (SD) of arithmetic means ($n = 9$). Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). 8-hour OMF and SMF exposure significantly suppressed bacterial growth. The cell growth values were analysed at 600 nm wavelength measuring the optical density (OD) by spectrophotometric reader after 8 h. OMF, oscillating magnetic field; SMF, static magnetic fields.

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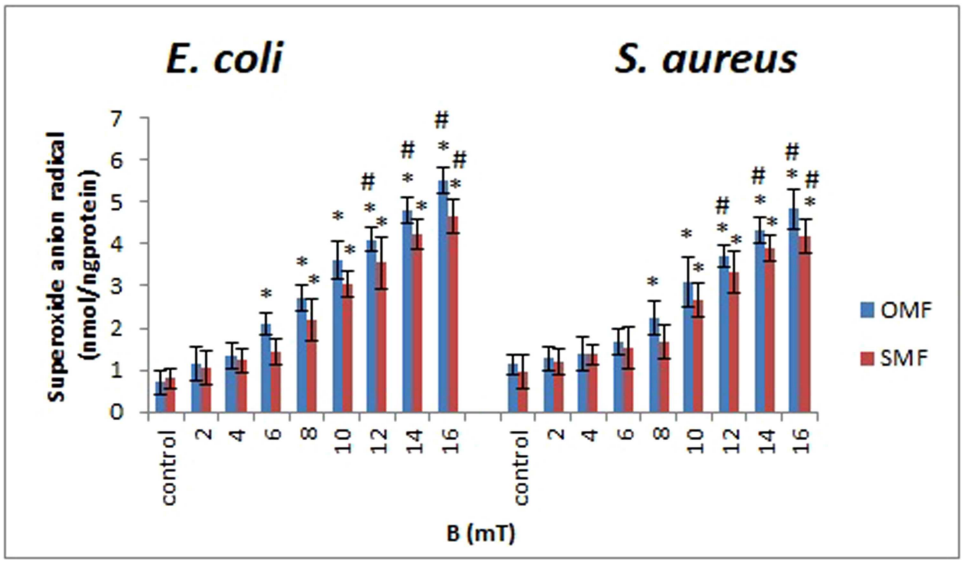

FIGURE 3 | OMF and SMF effects on superoxide anion radical level of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). 8-h OMF and SMF exposure significantly increased superoxide anion radical levels in both *E. coli* and *S. aureus*. OMF, oscillating magnetic field; SMF, static magnetic fields.

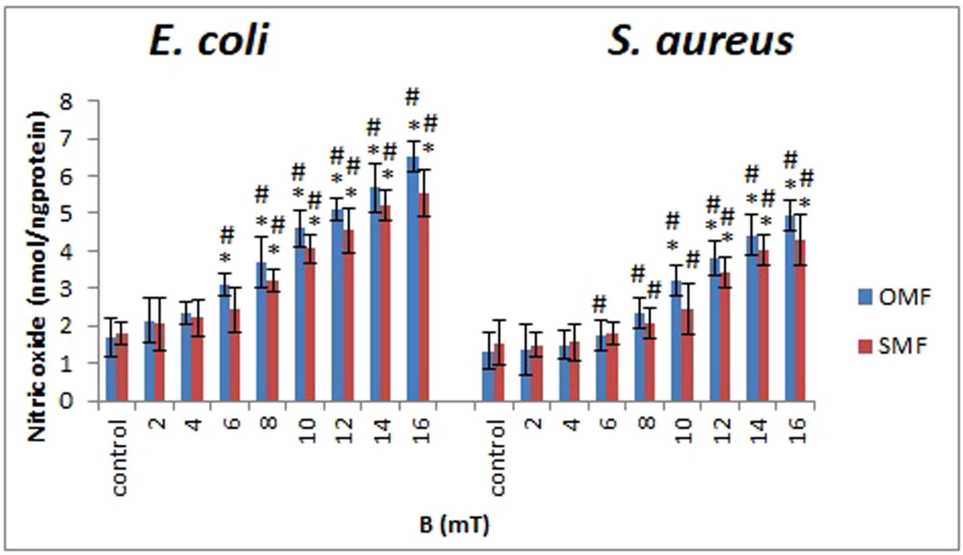

FIGURE 4 | OMF and SMF effects on nitric oxide level of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Exposure to the OMF and SMF resulted in a significant increase in nitric oxide levels in both *E. coli* and *S. aureus*. OMF, oscillating magnetic field; SMF, static magnetic fields.

There are findings that oxidative stress-mediated bactericidal activity of anti-microbial effective factors is a mechanism of anti-microbial agents (Ajiboye et al. 2016). Oxidative stress occurs due to changes in the activities of cellular antioxidant enzymes and free radical production (Kara et al. 2016; Baş et al. 2013). This study investigates the possible role of nitric oxide and superoxide anion radical, which are ROS, and oxidative stress in MF-mediated bactericidal activity against *S. aureus* and *E. coli*. Free radicals such as superoxide and NO are atoms, molecules or ions with unpaired electrons. These unpaired electrons are often extremely reactive and in biological systems they damage molecules such as enzymes and DNA (Apaydin et al. 2015). In this study, the applied magnetic field cause to increase the formation of free radicals (superoxide and NO) that damage cellular elements.

Superoxide dismutase and catalase are antioxidant enzymes that protect cells against free radicals (Baş and Ersoy 2020). The increased antioxidant enzyme level indicates that the cells are

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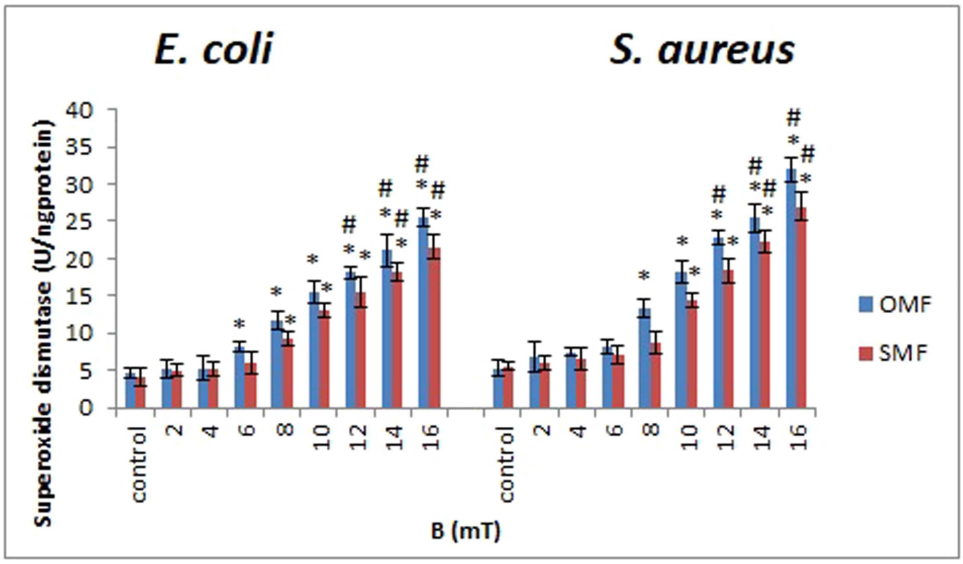

FIGURE 5 | OMF and SMF effects on superoxide dismutase activity of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Superoxide dismutase activity increased in response to both OMF and SMF. OMF, oscillating magnetic field; SMF, static magnetic fields.

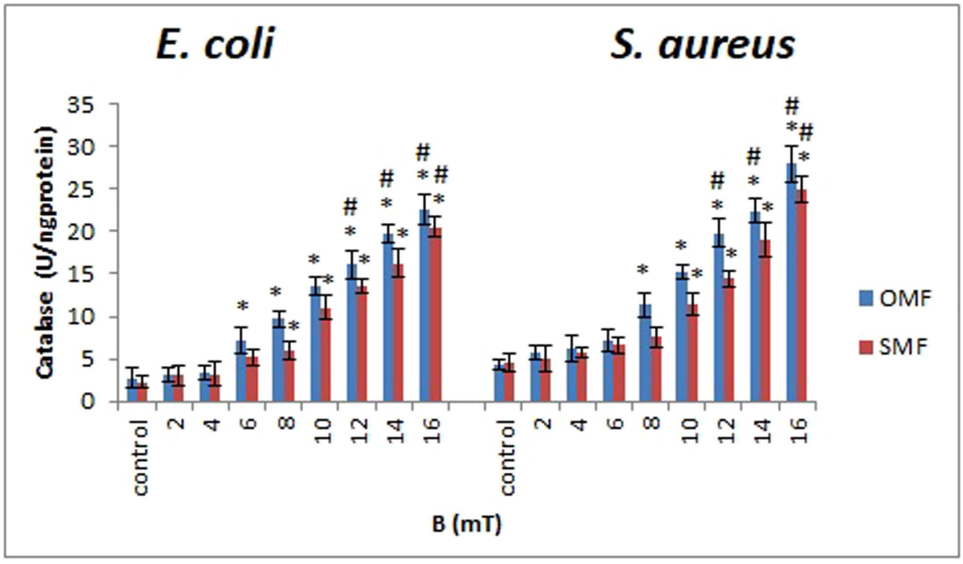

FIGURE 6 | OMF and SMF effects on catalase activity of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Catalase activity was significantly elevated by OMF and SMF. OMF, oscillating magnetic field; SMF, static magnetic fields.

on the defensive against the increased free radicals. In this study, an increase in antioxidant enzyme activities is observed in parallel with increasing superoxide and NO levels. GSH and GSSG participate in antioxidant defense reactions in cells. The changes in their levels indicate that the defense mechanisms against free radicals are activated, similar to the increase in antioxidant enzyme activities (Ajiboye et al. 2016). It is seen that the GSSG level increases and the GSH level decreases. These situations lead to an imbalance in the antioxidant defense system.

The magnitude of the observed changes supports their biological relevance. For example, ROS-related parameters such as superoxide and nitric oxide seriously increased compared with the control, which is comparable to increases reported in previous magnetobiological studies that demonstrated oxidative damage and reduced cell viability (Fojt et al. 2004; Gaafar et al. 2008; Konopacki and Rakoczy 2019; Nawrotek et al. 2014; Binhi and Rubin 2007). Therefore, these changes represent meaningful physiological stress responses, not merely statistical fluctuations. Moreover, the intensity-dependent trend was

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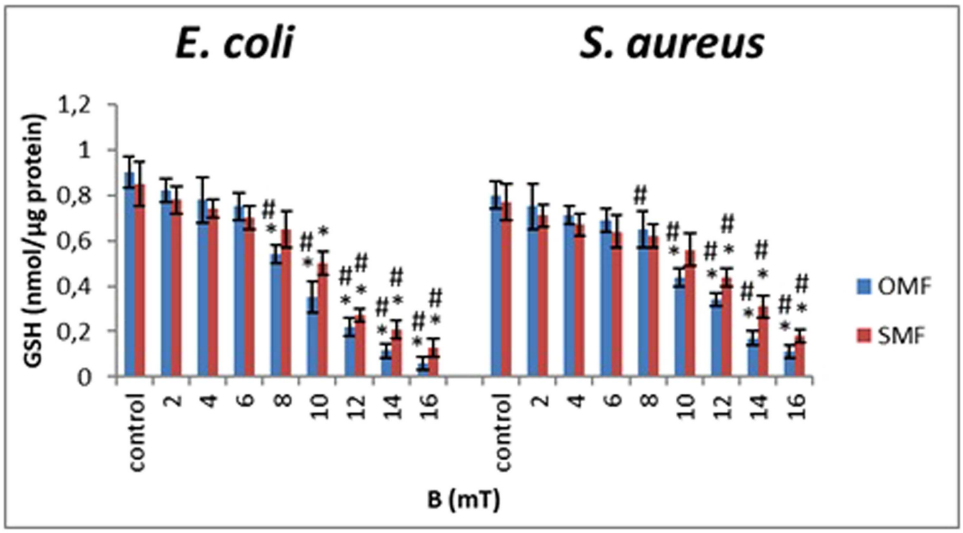

FIGURE 7 | OMF and SMF effects on glutathione reduced level of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Reduced glutathione levels decreased particularly under OMF and SMF exposure. OMF, oscillating magnetic field; SMF, static magnetic fields.

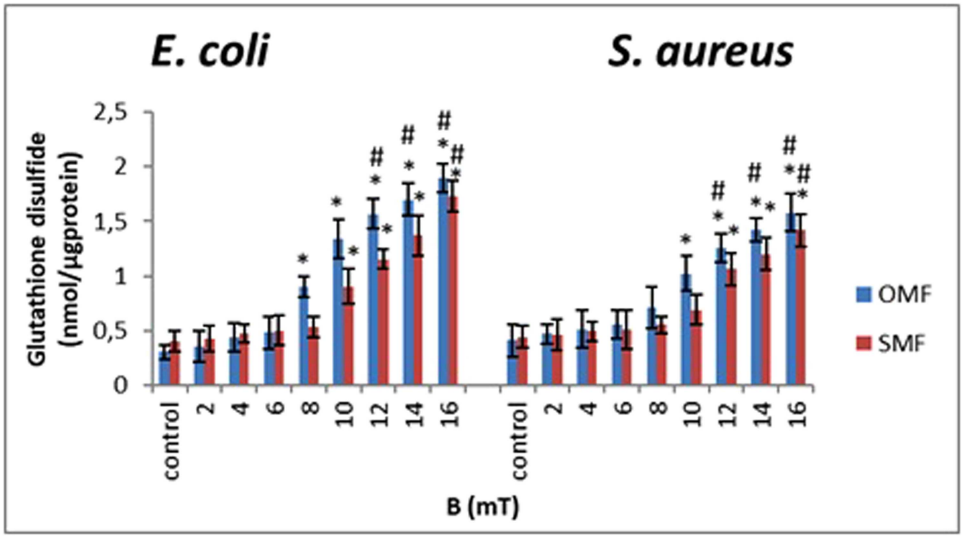

FIGURE 8 | OMF and SMF effects on glutathione disulfide level of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Glutathione disulfide (GSSG) levels increased in OMF and SMF-treated groups. OMF, oscillating magnetic field; SMF, static magnetic fields.

clearer under OMF exposure, supporting the idea that alternating fields induce stronger electron-spin perturbations and intracellular eddy currents than SMF fields.

There is normally a balance between the antioxidant defense system and free radical production. As it is known, disruption of this balance in favor of free radicals is called oxidative stress, and MDA level is one of the most important indicators of lipid peroxidation and therefore oxidative stress (Apaydin et al. 2015). For this reason, MDA level was also included among the investigated parameters. Increased MDA level means that cell membranes are affected by the magnetic field and antioxidant defense systems are inadequate. This can result in disruption of membrane organization and in functional loss and modification of proteins and DNA bases. DNA, the genetic material, is one of the most important structures damaged by oxidative stress. Breaks that occur in DNA structures are called DNA fragmentation and are indicators of DNA damage (Güven et al. 2015). Similar increase in fragmented DNA level in magnetic field treated cells is an indication of oxidative stress, which supports the previous study that MF could induced DNA damage (Pandir and Sahingoz 2014).

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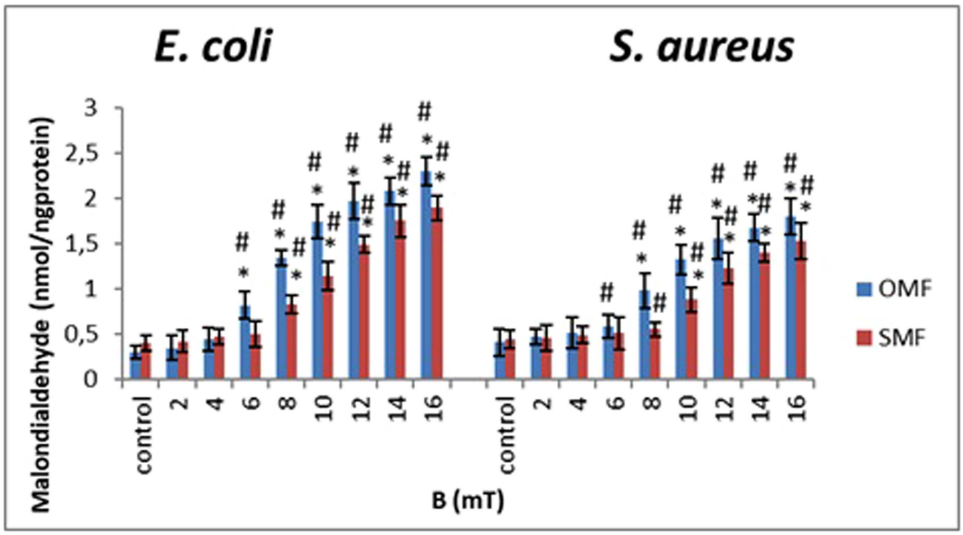

FIGURE 9 | OMF and SMF effects on malondialdehyde level of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). Malondialdehyde (MDA) concentrations were higher under OMF and SMF exposure. OMF, oscillating magnetic field; SMF, static magnetic fields.

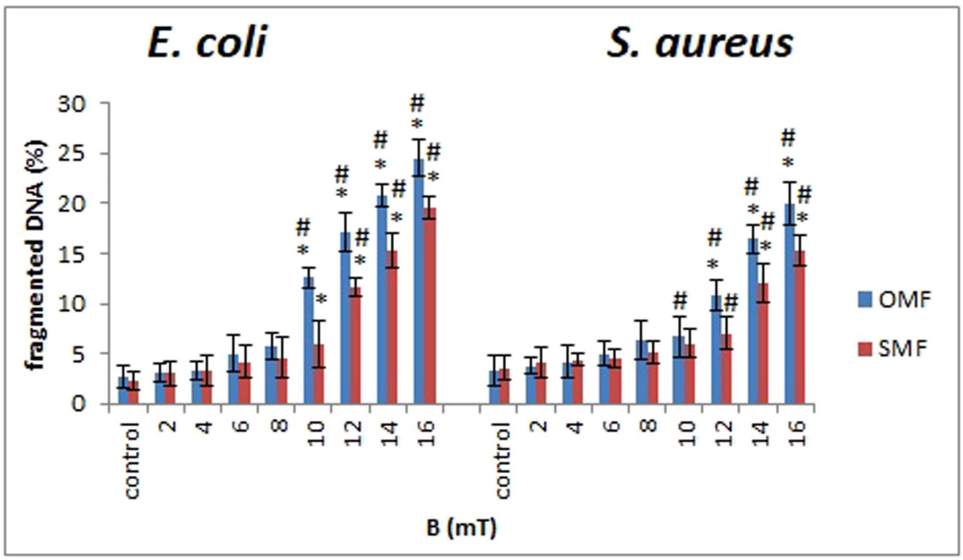

FIGURE 10 | OMF and SMF effects on fragmented DNA concentration of *Escherichia coli* and *Staphylococcus aureus*. Magnetic field intensities tested: 0 (control), 2, 4, 6, 8, 10, 12, 14, and 16 mT. Sign * means statistical significance at $p < 0.001$ between control and magnetic field-exposed groups. # means significant difference between *E. coli* and *S. aureus* at the same magnetic field intensity ($p < 0.05$). Findings were represented as the means ± standard deviation (SD) ($n = 9$). DNA fragmentation was significantly higher in bacteria exposed to OMF and SMF, revealing possible genotoxic effects of alternating magnetic exposure. OMF, oscillating magnetic field; SMF, static magnetic fields.

The intensity-dependent behavior observed across the tested magnetic field levels indicates that the biological effects are gradual. At 2–6 mT intensities, changes in oxidative stress and growth were minimal, while at 8–10 and 12–16 mT intensities pronounced differences appeared. The reduction in optical density reached maximum value at the highest MF exposure, and oxidative stress parameters (superoxide, NO, and MDA) increased relative to control. These data demonstrate that the magnetic field exerts a dose-dependent effect on bacterial viability and redox balance, supporting the progressive cellular stress.

A clear magnetic field–dependent trend was observed in both OMF and SMF exposures. For *E. coli*, statistically significant differences in oxidative stress markers and growth inhibition emerged at 6–8 mT for OMF and 8–10 mT for SMF fields, whereas for *S. aureus*, comparable effects appeared at slightly higher intensities (8–10 mT for OMF and 10–12 mT for SMF). This indicates that the onset of biological response depends not only on the field strength but also on the bacterial cell wall type and the nature of the magnetic field. The field-dependent response suggests the existence of a threshold intensity beyond which ROS generation and membrane damage become

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dominant. This observation supports the idea that OMF fields can induce cellular stress at lower thresholds than SMF fields, which may have implications for designing magnetic field–based antimicrobial applications.

If we interpret the results we have obtained so far, we can say that after a certain level, the magnetic field causes harmful effects on bacteria, and these harmful effects are realized by causing free radical production. As results it was revealed that for two groups of bacteria (Gram negative or Gram positive), different results can be obtained for SMF and OMF. Obtained data suggest that MF affects the cell membrane structure and that this effect is different for these two groups of bacteria due to differences in cell wall structure in Gram-negative and Gram-positive bacteria. And also, it was seen that the OMF is more effective on the bacterial cells as seen in figures. Because unpaired electron spins and orbits are affected more than SMF. The interspecies comparison further revealed that *E. coli* was more susceptible to magnetic field–induced oxidative stress and DNA damage than *S. aureus*. This difference became statistically significant at and above 8 mT (OMF) and 10 mT (SMF), likely reflecting differences in cell wall structure and permeability between Gram-negative and Gram-positive bacteria. Future studies combining magnetic field exposure with direct morphological analyses such as electron or fluorescence microscopy would provide further insight into the structural basis of the observed growth inhibition and oxidative damage.

Author Contributions

Hatice Baş: methodology, experimental analysis, formal analysis, writing – review and editing. Hatice **Kanbur Çavuş Hatice:** experimental analysis, writing – review and editing. **Nicola Morley** and **Recep Şahingöz:** supervision, writing – original draft.

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Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

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